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Lithium ion conductivity of polycrystalline perovskite $La_{0.67-x}Li_{3x}TiO_3$ with ordered and disordered arrangements of the A-site ions

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Abstract

Simple cubic perovskite $La_{0.67-x}Li_{3x}TiO_3$ ($x = 0.06$–0.15) with disordered arrangement of the A-site ions was prepared by quenching from 1350°C into liquid N_2, and tetragonal, doubled perovskite ($a = a_p$, $c \approx 2a_p$) with alternate arrangement of La-rich layers and Li-vacancy-rich layers along the *c*-axis, by annealing the quenched pellet at 800°C for 3 days. No extra lines due to superstructures other than the tetragonal polymorph were observed. Contrary to the expectation that the Li-vacancy-rich layers would be more favorable for fast conduction, the ordered samples (the parameter of order: $S \approx 0.55$–0.7) were found to have lower conductivity than the disordered samples in the range $x > 0.08$. A maximum bulk conductivity of 1.53×10^{-3} S cm^{-1} at 25°C, which is the best yet reported for known perovskite solid solutions, was found for the disordered sample with $x = 0.12$; a lower conductivity of 6.88×10^{-4} S cm^{-1}, was found for the ordered sample ($S \approx 0.6$). The lower conductivity is attributed to an increase in activation energy for ionic conduction that is associated with a decrease in the lattice parameter *a* of the primitive cell.

Keywords: Lithium ion; Perovskite; Ionic conductivity; Superstructure; Order–disorder transition

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1. Introduction

Fast lithium ion conducting solid electrolytes are essential to develop lithium solid-state batteries [1], which have some advantages such as no fear of leakage and combustion of electrolytes, flexibility in cell design and wide operating temperature. Considerable efforts have been directed toward search for oxide-based electrolytes [2] including γ_{II}-Li_3PO_4 solid solutions [3,4] and lithium analogues of NASICONs [5–7].

An exciting topic is the discovery of high lithium ion conductivity (ca. 10^{-3}–10^{-4} S cm^{-1} at room temperature) in titanate-based solid solutions with perovskite-related structures, such as $Ln_{0.67-x}Li_{3x}TiO_3$: Ln = La [8–12], (La,Nd) [12]; $Li_xSr_{1-x}Ta_xTi_{1-x}O_3$ [13]. Similar levels of ionic conductivity have been more recently found in tantalate-based perovskites, $Li_{2x}Sr_{1-2x}M(III)_{0.5-x}Ta_{0.5+x}O_3$ (M = Cr,Fe) [14]. They are one of the families with the highest

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conductivities so far reported for lithium ion conducting materials.

The recent studies [15] on the crystal structure of the $Ln_{0.67-x}Li_{3x}TiO_3$ (Ln = Pr,Nd) solid solutions have revealed that the lithium ion conduction occurs by an ion-vacancy mechanism through the interconnecting A-sites, making a detour round Ln^{3+}-occupied sites. In such a mechanism, site percolation can be expected to be one of determining factors for ionic conductivity. In fact, the composition dependence of conductivity has recently been explained in view of site percolation [16,17]. Three polymorphic forms were found in $La_{0.67-x}Li_{3x}TiO_3$: the high-temperature form with a cubic perovskite primitive cell ($a = a_p$) and the intermediate- and low-temperature forms with tetragonal, doubled perovskite unit cells ($a = a_p$, $c \approx 2a_p$) [18]. Very interestingly, the microdomain texture based on tetragonal, diagonal cells ($a = 2^{1/2}a_p$, $c = 2a_p$) has been observed in a composition close to $La_{0.5}Li_{0.5}TiO_3$ by high-resolution transmission electron microscopy [19].

The lithium ion conductivity found in $La_{0.67-x}Li_{3x}TiO_3$ ($x = 0.11$), $\sim 10^{-3}$ S cm^{-1} at 25°C [9,10], has been the highest so far reported for other oxide compounds. The X-ray pattern showed several weak superstructural peaks, indicating that La^{3+} ions, Li^{+} ions and vacancies were to a small extent arranged in the alternate La-rich and Li-vacancy-rich layers along the c-axis, with formation of a tetragonal superlattice doubled along the c-axis [20,21].

Anisotropic ionic conductivity has been recently observed in doubled perovskite $La_{0.59}Li_{0.27}TiO_{3.02}$ single crystal with the alternate arrangements: the ionic conductivities at 300 K parallel and perpendicular to the c-axis were 5.8×10^{-4} S cm^{-1} and 6.8×10^{-4} S cm^{-1}, respectively [22]. However, the conductivities were lower than that of the polycrystalline sample, and the single crystal was not well characterized in terms of quality, degree of order and crystallographic anisotropy including microstructure.

It is therefore essential and very attractive to examine conductivity in the polymorphic forms with different arrangements of the A-site ions. In this study, we prepare the cubic perovskite solid solution with a disordered arrangement and the tetragonal form with the alternate arrangement along the c-axis, and make a comparison of the conductivity of the disordered and ordered polymorphs.

2. Experimental

The samples were prepared by a conventional solid-state reaction. Reagent grade La_2O_3, Li_2CO_3, and TiO_2 (Wako Pure Chemical Ind.) were dried prior to use at 140°C for 5 h in a vacuum, and the required amounts of the dried reagents were thoroughly mixed in an agate mortar with a pestle. The lithium content was adjusted by adding an excess of Li_2CO_3, so as not to deviate from the intended stoichiometry due to the evaporation of lithia during sintering. The mixtures were heated initially at 650°C for 2 h to expel CO_2 gas and calcined at 800°C for 12 h on Au-boats. The reground products were cold-pressed into pellets (12 mm in diameter, 1–2 mm in thickness) at 250 MPa, which were sintered at 1350°C on Pt-boats for 1 h in a covered MgO crucible. The quenched samples were prepared by quenching the pellets from sintering temperature into liquid nitrogen. To avoid differences in conductivity among samples, we lapped off the gold electrodes of the quenched pellets very carefully after ac impedance measurements. The annealed samples were prepared by annealing the same quenched pellets at 800°C for 3 days, followed by subsequent quenching from annealing temperature. Typically, sintered pellets had a porosity of 8–18%, which generally decreased with increasing x. When their lithium contents, determined by flame emission analysis, were within 97–103% of the intended lithium content, the sintered samples were used for subsequent analyses. Analysis of the crystalline products was made by powder X-ray diffraction (XRD) with a Rigaku RAD-C system (Ni-filtered Cu Kα, Si internal standard) and the Cell-series programs [23]. The parameter of order (S) is defined for the alternate arrangement in the tetragonal polymorph as $S = [R(\text{La-rich}) - R(\text{dis})]/[1 - R(\text{dis})]$, where R(La-rich) and R(dis) are the occupancies of the A-sites by La^{3+} ions in the La-rich layers of the ordered form and in the (001) planes of the disordered form, respectively. The S values for the samples were determined as follows. The calculated relative intensities, $I(001)/I(110)$, were computed for the tetragonal polymorph with different x values at 0.1 intervals of S by using the simulating mode of Izumi's Rietveld program [24] and plotted against S for the x values of the samples. The S values for the samples were determined by using the plots and the observed relative intensities.

Prior to ac-conductivity measurements, the opposite faces of the sintered pellets were polished with successive lapping films, and gold electrodes were attached by dc sputtering onto the polished faces. The conductivity cell was placed between two outer gold electrodes of a conductivity jig, which was put in a Pyrex tube. The tube was immersed in a thermostatted ethylene glycol–water bath with a programmable temperature PID-controller, whereby the temperature of the cell was controlled within ±0.10°C in a range of −60°C–60°C. Below −60°C, the temperature was controlled within ±0.20°C by a laboratory-made cryostat using a rubber heater and liquid nitrogen. The cell was kept for about one hour at each temperature to achieve the temperature equilibrium prior to measurements. The ac impedance data were collected under a dry Ar gas flow with an impedance analyzer (YHP 4192A) over a frequency range, $f = 5$ Hz–13 MHz, and analyzed using the complex impedance or admittance diagrams (i.e., Z'–Z'' and Y'–Y'' plots), or the complex impedance and electric modulus spectra (i.e., Z''–f and M''–f plots), whichever was most appropriate [25].

3. Results and discussion

Fig. 1a and b, shows the powder XRD patterns of the quenched and annealed samples with $x = 0.12$, along with the simulated pattern, Fig. 1c, of the annealed sample. All the reflections of the quenched sample were indexed in a simple cubic perovskite unit cell (S.G: Pm$\bar{3}$m; $a = a_p$); This indicates that the La ions, Li ions and vacancies are randomly distributed over the A-sites. For the annealed sample, the reflections including additional superstructural lines were indexed well in a tetragonal, doubled cell (P4/mmm; $a = a_p$, $c = 2a_p + \delta$) with alternate arrangement of La-rich layers and Li-vacancy-rich layers along the c-axis. The pattern simulated for the order parameter $S = 0.6$ fitted that of the annealed sample well, as shown in Fig. 1c. In our preparation conditions, we have never observed the superstructural lines due to the diagonal cell [19] and the cubic, doubled cell [9], probably because of the low calcinating temperature.

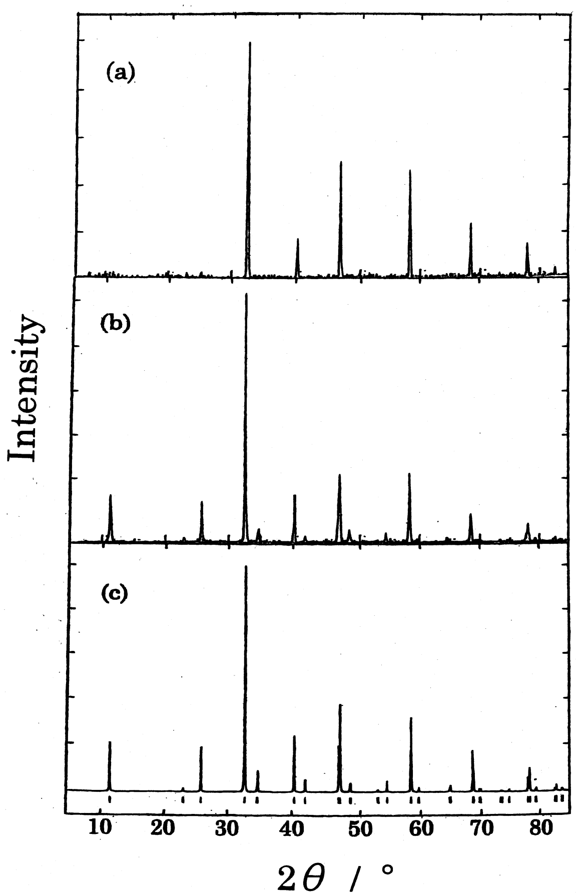

Fig. 1. Powder XRD patterns of the quenched and annealed samples with $x = 0.12$ and the simulated pattern of the annealing sample; (a) quenched, (b) annealed, (c) simulated.

Fig. 2 shows the lattice parameters for the quenched and annealed samples as a function of x. The lattice parameter a for the cubic form decreased with increasing x. In a range of $x < \sim 0.08$, the cubic form was not quenched and very weak superstructural lines remained in the XRD patterns. On the other hand, all the annealed samples adopted the tetragonal, doubled cell. The parameter S changed form 0.53 for $x = 0.15$ to 0.7 for $x = 0.06$. The lattice parameter c decreased by about 2 pm with increasing x, while the parameter a remained almost constant. As a result, the parameter $V^{1/3}$ decreased roughly parallel to the slope for the cubic parameter with increasing x. The alternate arrangement thus gives rise to tetragonal distortion and contraction in cell volume and in the a-axis of the primitive cell.

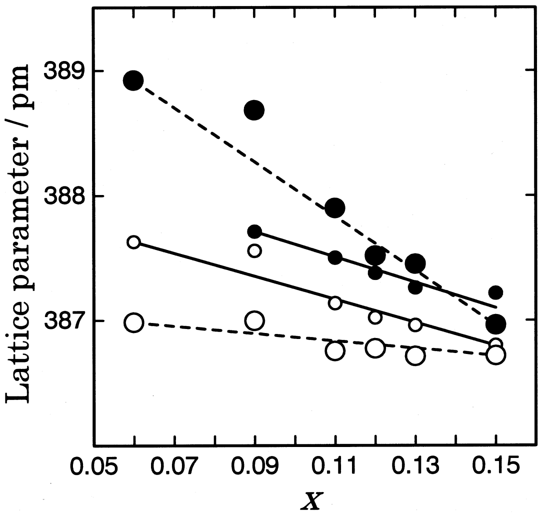

Fig. 2. Lattice parameters for the simple cubic form and the tetragonal, ordered form as a function of *x*. •: *a* for the simple cubic cell; ∘: $V^{1/3}$ (*V*: volume of the subcell of the tetragonal form); ○: *a* and ●: *c*/2 for the tetragonal cell.

Fig. 3 shows a comparison of bulk conductivities at 25°C for the quenched and annealed samples. The quenched samples had higher conductivity than the annealed samples in a range of $x > \sim 0.08$, while the opposite occurred in a range of $x < 0.08$. The dome-shape dependence of conductivity became dull for the annealed samples, partly because the low concentration of La^{3+} ion in the Li-vacancy-rich layers suppresses site percolation effects [16,17] on the low *x* value side. A maximum conductivity was found at the same composition of $x = 0.12$ for both samples: 1.53×10^{-3} S cm^{-1} for the quenched sample, which is the best yet reported for known perovskite solid solutions, and 6.88×10^{-4} S cm^{-1} for the annealed sample ($S \approx 0.6$). The quenched sample with $x = 0.11$ was subjected to quenching-annealing cycles. The variations in conductivity during the cycles, shown in Fig. 4, confirm more clearly that the quenched sample has higher conductivity, and demonstrate that the order–disorder transition is reversible.

Fig. 5 shows the Arrhenius plots of the bulk conductivities the disordered and ordered samples with $x = 0.12$. The plots are linear, indicating that no phase transition occurs in either, in the temperature range, -100°C–60°C. Table 1 tabulates the conductivity data for both samples. The activation energy for the disordered sample was 31.6 kJ mole^{-1}, which was slightly smaller than 33.9 kJ mole^{-1} for the ordered sample. In view of the correlation between activation energy and lattice parameters in other lithium ion conducting perov-

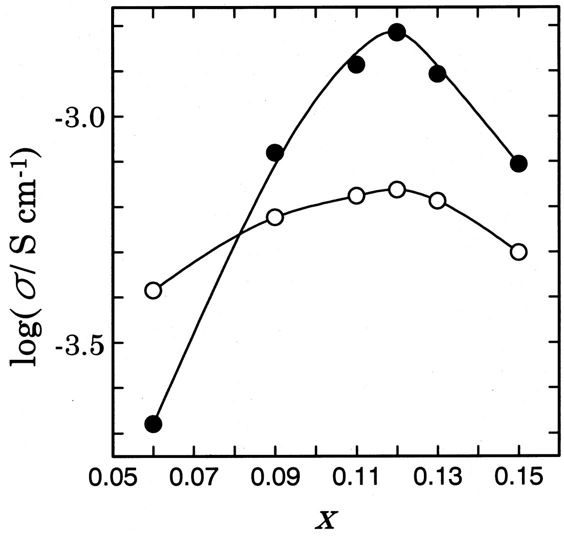

Fig. 3. Bulk conductivities at 25°C for the quenched and annealed samples; ●: quenched; ○: annealed.

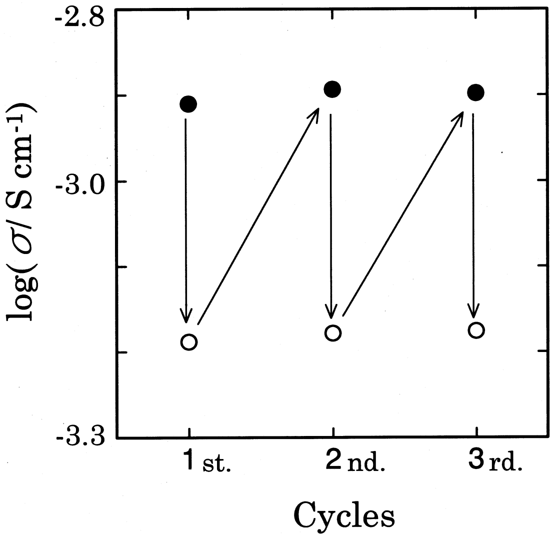

Fig. 4. Variations of ionic conductivity for the sample with $x = 0.11$ during quenching–annealing cycles; ●: quenching; ○: annealing.

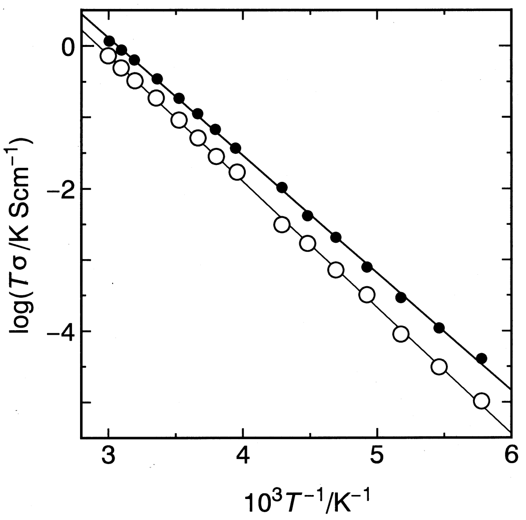

Fig. 5. Arrhenius plots of the bulk conductivities for the disordered and ordered samples with $x = 0.12$; ●: quenched; ○: annealed.

skites [10,12], it is reasonable that the increase in activation energy is caused by contraction in the a-axis of the primitive cell. The conductivity prefactor for the ordered sample was larger than that for the quenched sample. Table 2 lists the calculated molar fractions of La^{3+}, Li^{+} and vacancy in the (001) plane of the disordered form and in the Li-vacancy-rich and La-rich layers of the tetragonal, ordered form. The increase in the prefactor is, however, rather smaller than expected from the differences in the calculated fractions in the (001) plane of the disordered form and in the Li-vacancy-rich layer of the tetragonal, ordered form.

Fig. 6 shows the M''–f plots (a) and the normalized M''–f plots (b) at −40°C for the disordered and ordered samples with $x = 0.12$. Theoretically, the frequency, f_{max}, of the peak maximum of M'' is given by the equation, $f_{max} = \sigma/e_0\varepsilon'$ [25], where σ, e_0 and ε' are ionic conductivity and the permittivities of free space and the ionic conductor. In fact, the frequency f_{max} for the quenched sample with higher conductivity was higher than that of the annealed sample. The magnitudes of full width half maximum (W_h) of both peaks were rather large compared with the theoretical value (1.14 decades) of the Debye peak, which would expected for an equivalent circuit comprising a single, parallel RC element. The normalized plots demonstrate that the annealing slightly increased the W_h value from 2.1 decades for the quenched sample to 2.25 for the annealed sample, and that the broadening occurs on the low frequency side of the peak. At least three explanations can be advanced for it: (i) the random orientation of grains with anisotropic conductivity [25], (ii) an increase in the variety of lithium sites due to the superstructure, and (iii) an increase in coupling motion between

Table 1
Conductivity data for the cubic, disordered form and the tetragonal, ordered form ($S \approx 0.6$) with $x = 0.12$

Phase	Bulk conductivity (25°C) (S cm^{-1})	Prefactor σ_0 (K S cm^{-1})	Activation energy E_{Li} (kJ mole^{-1})
Cubic, disordered	1.53×10^{-3}	1.5×10^{5}	31.6
Tetragonal, ordered	6.88×10^{-4}	1.8×10^{5}	33.9

Table 2
Calculated molar fractions of the La^{3+} ion, Li^{+} ion and vacancy in the (001) plane of the disordered form and in the La-rich and Li-vacancy-rich layers of the tetragonal, ordered form with $S = 0.6$

	Cubic, disordered form (001) plane	Tetragonal, ordered form ($S = 0.6$)	
		La-rich layer	Li-vacancy-rich layer
La^{3+} ion	0.550	0.820	0.280
Li^{+} ion	0.360	(0.144)[b]	(0.576)[b]
Vacancy	0.090	(0.036)[b]	(0.144)[b]

[a] The composition was $x = 0.12$.
[b] Assuming random distribution of lithium ions and vacancies to both layers.

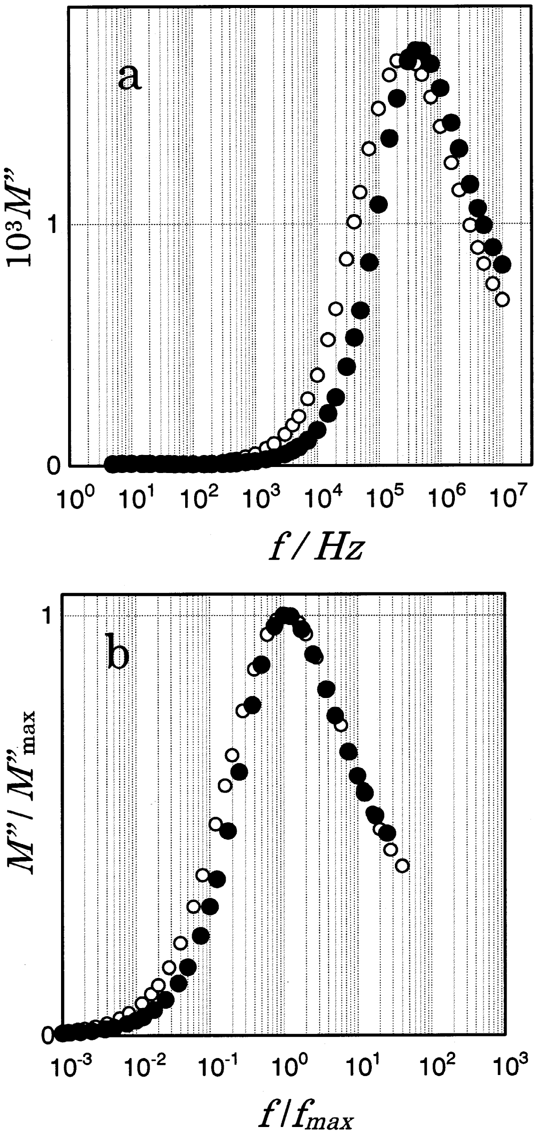

Fig. 6. M''–f plots (a) and normalized M''–f plots (b) at $-40°C$ for the quenched and annealed samples with $x = 0.12$; ●: quenched; ○: annealed.

lithium ions in the Li-vacancy-rich layers [26]. At present, we can not judge their relative contributions to the width.

The important prediction from the results described above is that the anisotropic conductivity in the ordered perovskite would not be as high as that of β-alumina. We however needs further sophisticated studies to confirm this.

4. Conclusions

(1) Contrary to the expectation that the Li-vacancy-rich layers would be more favorable for fast conduction, the tetragonal form (order parameter $S \approx 0.5$–0.7) with the alternate arrangement of La-rich layers and Li-vacancy-rich layers along the *c*-axis was found to have lower conductivity than the simple cubic form with disordered arrangements of the A-site ions in polycrystalline $La_{0.67-x}Li_{3x}TiO_3$ ($x > 0.08$).

(2) The lower conductivity in the tetragonal form is attributed to an increase in activation energy for ionic conduction that is associated with contraction in the *a*-axis of the primitive cell.

(3) The simple cubic form with $x = 0.12$ had a maximum bulk conductivity of 1.53×10^{-3} S cm^{-1} (25°C), which is the best yet reported for known perovskite solid solutions. A lower conductivity of 6.88×10^{-4} S cm^{-1} was found in the tetragonal form ($S \approx 0.6$).

(4) No extra reflections due to superstructures other than the tetragonal polymorph were observed in the annealed samples.

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